

NEF in one line

Principle 1 (representation): nonlinear encoding $x \rightarrow a$, linear decoding $a \rightarrow \hat{x}$. **Principle 2 (transformation):** connection weights compute $y = f(x)$.

$$\underbrace{J_i = \alpha_i \langle e_i, x \rangle + J_i^{\text{bias}}}_{\text{encode}} \rightarrow \underbrace{a_i = G(J_i)}_{\text{neuron}} \rightarrow \underbrace{\hat{x} = \sum_i d_i a_i}_{\text{decode}}$$

Neuron models (L2)

ReLU $G(J) = \max(J, 0)$.

LIF subthreshold (normalized: rest 0, $v_{\text{th}} = 1$)

$$\frac{dv}{dt} = \frac{1}{\tau_{RC}}(J - v), \quad v(t) = J + (v_0 - J)e^{-t/\tau_{RC}}.$$

Spike when $v \geq 1$; reset $v = 0$, hold for τ_{ref} ; clamp $v \geq 0$.

Time to threshold (from $v_0 = 0$, needs $J > 1$)

$$t_{\text{th}} = -\tau_{RC} \ln(1 - \frac{1}{J}), \quad \text{ISI} = \tau_{\text{ref}} + t_{\text{th}}.$$

LIF rate

$$G(J) = \frac{1}{\tau_{\text{ref}} - \tau_{RC} \ln(1 - 1/J)} \quad (J > 1), \quad 0 \text{ otherwise.}$$

Inverse $J(a) = \frac{1}{1 - \exp((\tau_{\text{ref}} - 1/a)/\tau_{RC})}$. $\triangleright$ Max possible rate

$1/\tau_{\text{ref}}$ (500 Hz for 2 ms). LIF is concave, steep just above threshold; ReLU is a straight line.

Euler step $v_{k+1} = v_k + \frac{\Delta t}{\tau_{RC}}(J - v_k) = (1 - \frac{\Delta t}{\tau_{RC}})v_k + \frac{\Delta t}{\tau_{RC}}J$. From $v_0 = 0$: $v_k = J(1 - (1 - \frac{\Delta t}{\tau_{RC}})^k)$. Spikes snap to the grid, so the ISI rounds up and high rates are most affected.

Gain & bias from a_{max} , ξ (L2–3)

Conditions: $J(\text{at } ex = r) = J_{\text{max}} = G^{-1}(a_{\text{max}})$, $J(\text{at } ex = \xi) = J_{\text{th}}$.

$$\alpha = \frac{J_{\text{max}} - J_{\text{th}}}{r - \xi}, \quad J^{\text{bias}} = J_{\text{th}} - \alpha\xi.$$

- ReLU: $J_{\text{th}} = 0$, $J_{\text{max}} = a_{\text{max}} \Rightarrow \alpha = \frac{a_{\text{max}}}{r - \xi}$, $J^{\text{bias}} = -\alpha\xi$.
- LIF: $J_{\text{th}} = 1$, $J_{\text{max}} = J(a_{\text{max}})$ from the inverse above.
- Radius r : same as using x/r with radius 1; gain becomes $r\alpha$.
- Intercept in x : neuron with $e = +1$ fires for $x > \xi$; with $e = -1$ fires for $x < -\xi$.

$\triangleright$ Two-rate spec (e.g. 40 Hz at $x = 0$, 150 Hz at $x = 1$): $J^{\text{bias}} = J(40)$, $\alpha = J(150) - J(40)$.

Vectors & tuning (L3)

$J = \alpha \langle e, x \rangle + J^{\text{bias}}$, with $\|e\| = 1$ (preferred direction).

- Threshold is the line/plane $\langle e, x \rangle = \xi$; the half-space beyond it fires.
- On the unit circle $x = (\cos \theta, \sin \theta)$: $\langle e, x \rangle = \cos(\theta - \theta_e)$. The tuning curve is “cosine-like” but rectified and LIF-warped.
- Random uniform encoders: angle $\phi \sim U[0, 2\pi)$, $e = (\cos \phi, \sin \phi)$.

Least-squares decoders (L3)

Shapes (n neurons, m or N samples, d dims): $A \in \mathbb{R}^{n \times N}$, $X \in \mathbb{R}^{d \times N}$, $D \in \mathbb{R}^{d \times n}$, $E \in \mathbb{R}^{n \times d}$.

Objective $\min_D \|X - DA\|_F^2 + N\sigma^2 \|D\|_F^2$.

$$D^\top = (AA^\top + N\sigma^2 I)^{-1}AX^\top$$

Scalar: $d = (AA^\top + N\sigma^2 I)^{-1}Ax$, $\hat{x} = d^\top A$.

- $\sigma = 0$: normal equations $(AA^\top)d = Ax$.

- Noise σ is often specified as a fraction of max A (e.g. $0.2 \max A$).
- Fit on *clean* A ; noise enters only through the $N\sigma^2 I$ term.
- Regularization shrinks d : slightly worse on clean data, much better on noisy data.
- Units: A in Hz $\Rightarrow d$ in s.

2×2 inverse $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. $\triangleright$ Symmetric pair with antisymmetric RHS: try $d = (\delta, -\delta)$, then $\delta = \text{rhs}_1/(a-b)$.

Error measures & scaling (L3)

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_k (x_k - \hat{x}_k)^2}, \quad \text{vector: } \sqrt{\frac{1}{Nd} \sum_{k,j} (\hat{x}_{jk} - x_{jk})^2}$$

$$E = E_{\text{dist}} + E_{\text{noise}}, \quad E_{\text{noise}} = \sigma^2 \sum_i d_i^2,$$

$$E_{\text{dist}} = \frac{1}{N} \sum_k (x_k - \hat{x}_k^{\text{clean}})^2.$$

- $E_{\text{noise}} \propto 1/n$ (doubling n halves it; log-log slope -1).
- $E_{\text{dist}} \propto 1/n^2$ (doubling n quarters it; slope -2).
- Fixed decoders: $\sigma \rightarrow \sigma/10$ gives $E_{\text{noise}} \rightarrow E_{\text{noise}}/100$.
- RMSE then scales as $1/\sqrt{n}$ in the noise-limited regime.

Population vector $\hat{x} \approx E^\top a$ (encoders as decoders). Directions are roughly right for broad, uniform populations, but magnitudes are wrong. Fitted D is better overall. $\triangleright$ Normalize columns to unit length to compare *direction* only.

Spike trains (L4)

$$a_i(t) = \sum_k \delta(t - t_i^k), \quad \int \delta = 1.$$

Discrete: a spike is a sample of height $1/\Delta t$ (so area = 1). $\triangleright$ If spikes are stored as 1s, a factor $1/\Delta t$ must appear somewhere for filtered trains to be in Hz. Sanity check: steady 150 Hz gives filtered ≈ 150 .

Rate decoders on spikes $\hat{x}(t) = ((Da) * h)(t) = D(a * h)(t)$ (linear, so the filter can go before or after D).

Two opposite neurons $d_- = -d_+ \Rightarrow \hat{x} = (r * h)$, with $r = a_+ - a_-$ (signed spikes).

Fourier essentials (L4)

$$X(\omega) = \int x(t)e^{-i\omega t} dt, \quad \omega = 2\pi f.$$

- Real signal $\Leftrightarrow X(-\omega) = \overline{X(\omega)}$; then a pair gives $2\text{Re}[Xe^{i\omega t}]$.
- $X = a + ib$: pair = $2a \cos \omega t - 2b \sin \omega t$; magnitude $|X|$, phase arg X .
- Convolution theorem: $\mathcal{F}[f * g] = FG$.
- Linear: $\mathcal{F}[af + bg] = aF + bG$.
- Parseval (series): $\frac{1}{T} \int_0^T x^2 dt = \sum_k |X_k|^2 \Rightarrow \text{RMS}$.
- A sinusoid of amplitude A has mean square $A^2/2$.

Discrete grid $N = T/\Delta t$, $\Delta\omega = 2\pi/T$ (i.e. $\Delta f = 1/T$), Nyquist $f = 1/(2\Delta t)$.

Band-limited white noise Random X at $|f| \leq \text{limit}$, zero above; enforce conjugate symmetry; inverse FFT; rescale to the target RMS (rescaling does not change spectral shape). $\triangleright$ Higher limit means faster wiggles at the same RMS. One spectrum is ragged; the average over many is flat up to the limit.

Filters (L4)

Convolution $(f * h)(t) = \int f(t-t')h(t') dt'$; on spikes, $\sum_k h(t - t_k)$.

Gaussian $h = c^{-1}e^{-t^2/\sigma^2}$, unit area; acausal, low-pass.

Optimal filter minimize $\langle |X - RH|^2 \rangle$ per frequency:

$$H(\omega) = \frac{\langle X \overline{R} \rangle}{\langle |R|^2 \rangle} \quad \text{windowed: } \frac{(X \overline{R}) * W}{|R|^2 * W}$$

- Derivation: expand, set $\partial E / \partial \bar{H} = 0$.
- Where the signal has no power, $H = 0 \Rightarrow$ low-pass.
- $|R|$ is broadband (deltas have flat spectra); $|\hat{X}| = |H||R|$.
- Windowing in ω = multiplying $h(t)$ by a time window, giving a short, smooth h .
- **Acausal** ($h(t < 0) \neq 0$): fine offline, not biological.

PSC / synaptic filter (causal)

$$h(t) = \frac{t^n e^{-t/\tau}}{n! \tau^{n+1}}, \quad t \geq 0; \quad H(\omega) = \frac{1}{(1 + i\omega\tau)^{n+1}}$$

Peak $t = n\tau$; height $1/\tau, 1/(e\tau), 2e^{-2}/\tau$ ($n = 0, 1, 2$)
 Mean delay $(n+1)\tau$
 Cutoff ($n = 0$) $f_c = 1/(2\pi\tau), |H| = 1/\sqrt{1 + \omega^2\tau^2}$

- $\uparrow n$: smoother, more delay. $\uparrow \tau$: less spike noise, more lag, attenuates fast signals.
- $n = 0$ by hand: each spike adds a jump of d/τ ; between events everything decays by $e^{-\Delta t/\tau}$.
- Typical τ (lecture 4): excitatory ≈ 5 ms, inhibitory GABA_A ≈ 10 ms.

▷ Slow signals relative to τ decode better: lag is a smaller fraction of their timescale.

Transformations (L5)

Function decoder replace targets X by $Y = f(X)$:

$$(D^f)^T = (AA^T + N\sigma^2 I)^{-1} AY^T$$

- Linear in the target: $D^{aF+bG} = aD^F + bD^G$.
- Linear $f = Fx$: $D^f = FD$ exactly. Affine $2x+1$: $D^f = 2D + D^1$ (need a constant decoder; $2D$ alone gives only $2x$).
- Smooth f decode well; jagged or discontinuous f decode badly. Error falls with n .

Weights

$$J_i = \sum_j w_{ij} a_j + J_i^{\text{bias}}, \quad w_{ij} = \alpha_i \langle e_i, d_j^f \rangle, \quad W = ED^f$$

(E includes the gains). W has rank $\leq d$.

- Cost: full W is $\mathcal{O}(nm)$; decode-then-encode is $\mathcal{O}(d(n+m))$ (e.g. 40,000 vs 400 for $n = m = 200, d = 1$).
- Dale's principle: each pre-neuron's outgoing weights should share one sign; ED usually violates it.
- $w > 0$ excitatory, $w < 0$ inhibitory, $w = 0$ no synapse.

Multiple inputs currents add:

$$J_i = \alpha_i \langle e_i, D^f a_1 + D^g a_2 \rangle + J_i^{\text{bias}} \approx \alpha_i \langle e_i, f(x) + g(y) \rangle + J_i^{\text{bias}}$$

- Separate pre-populations only give $f(x) + g(y)$.
- Nonlinear in both (e.g. xy): use a 2D population representing (x, y) and decode $f(x, y)$.
- Keep the output inside the target's radius: bound $|z|$ or estimate $\text{RMS}(z) = \sqrt{\sum_k c_k^2 \text{Var}(x_k)}$ (independent inputs); otherwise enlarge the radius.

Deviations in a chain lag of about τ per synaptic stage (an n -stage chain lags about $n\tau$); start-up transient (filters start at 0); spike ripple; edge distortion near $\pm r$; saturation beyond r .

Recipe (1) define encoders for input and output populations; (2) write $y = f(x)$; (3) solve D^f on the input population; (4) substitute into the output encoding, $J = \alpha \langle e, D^f a \rangle + J^{\text{bias}}$.

Handy numbers ($\tau_{\text{ref}} = 2$ ms, $\tau_{RC} = 20$ ms)

a (Hz)	$J(a)$	note
40	1.463	$e^{-1.15} = 0.3166$
100	3.033	$e^{-0.4} = 0.6703$
150	4.805	$e^{-0.2333} = 0.7919$
200	7.179	$e^{-0.15} = 0.8607$

Euler with $\Delta t = 1$ ms: $\Delta t/\tau_{RC} = 0.05$, $\ln 0.95 = -0.05129$.
 $e^{-1} = 0.368$, $e^{-2} = 0.135$, $e^{-3} = 0.0498$, $\sin \frac{\pi}{4} = \cos \frac{\pi}{4} = 0.7071$.